Summary in Graph

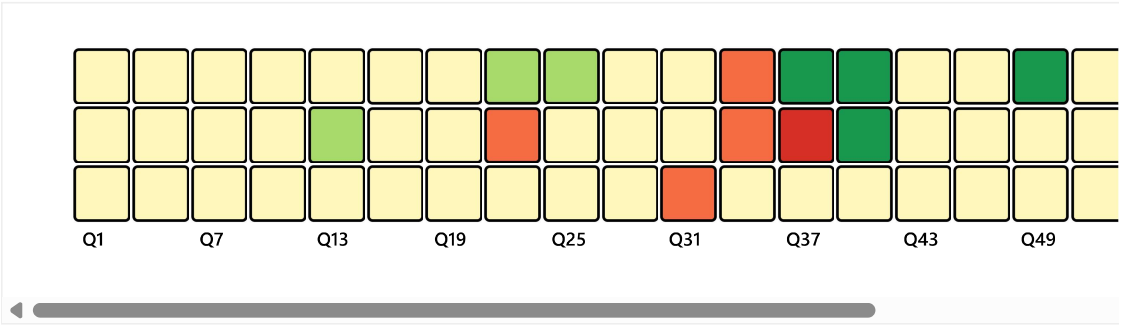
Exam Summary (GO Classes DA 2026 | All India Mock Test 3)

Qs. Attempted:	18 (7 + 11)	Correct Marks:	17 (3 + 14)
Correct Attempts:	10 (3 + 7)	Penalty Marks:	2 (1.33 + 0.67)
Incorrect Attempts:	8 (4 + 4)	Resultant Marks:	15 (1.67 + 13.33)

Total Questions:	65 (30 + 35)
Total Marks:	100 (30 + 70)
Exam Duration:	180 Minutes
Time Taken:	180 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (27) out of 270 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Aptitude

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

Which of the following is the value of $\sqrt{\log_2 6 + \log_3 6}$?

A. 1

B. $\sqrt{\log_5 6}$

C. 2

D. $\sqrt{\log_2 3} + \sqrt{\log_3 2}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 04min 08sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

David drives from his home to the airport to catch a flight. He drives 35 miles in the first hour, but realizes that he will be 1 hour late if he continues at this speed. He increases his speed by 15 miles per hour for the rest of the way to the airport and arrives 30 minutes early. How many miles is the airport from his home?

A. 140

B. 175

C. 210

D. 245

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

A positive number is increased by 60%. By what percentage should the result be decreased to return to the original value?

A. 57.5%

B. 40%

C. 62.5%

D. 37.5%

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Quantitative Aptitude

The average of a, b and c is 16. The average of c, d and e is 26. The average of $a, b, c, d,$ and e is 20. The value of c is

A. 10

B. 20

C. 21

D. 26

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Verbal Aptitude

Inhaling the smoke from a burning _____ could _____ you quickly.

A. tire / tier

B. tire / tyre

C. tyre / tire

D. tyre / tier

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Two distinct numbers are selected from the set $\{1, 2, 3, 4, \ldots, 36, 37\}$ so that the sum of the remaining 35 numbers is the product of these two numbers. What is the difference of these two numbers?

A. 7

B. 8

C. 9

D. 10

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

A factory makes chocolate bars. Five boxes, labelled V, W, X, Y, Z, are each packed with 20 bars. Each of the bars in three of the boxes has a mass of 100g. Each of the bars in the other two boxes has a mass of 90g. One bar is taken from box V, two bars are taken from box W, four bars are taken from box X, eight bars are taken from box Y, and sixteen bars are taken from box Z. The total mass of these bars taken from the boxes is 2920g. The boxes containing the 90g bars are labelled

A. V and W

B. W and Z

C. X and Y

D. V and Z

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Quantitative Aptitude

Jack wants to bike from his house to Jill's house, which is located three blocks east and two blocks north of Jack's house. After biking each block, Jack can continue either east or north, but he needs to avoid a dangerous intersection one block east and one block north of his house. In how many ways can he reach Jill's house by biking a total of five blocks?

A. 4

B. 5

C. 6

D. 8

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Analytical Aptitude

Twenty-five cards are randomly arranged in a grid, as shown.

0	0	1	1	0
1	0	0	1	0
1	1	0	1	0
0	0	1	0	0
1	0	1	0	1

Five of these cards have a 0 on one side and a 1 on the other side. The remaining twenty cards either have a 0 on both sides or a 1 on both sides. Loron chooses one row or one column and flips over each of the five cards in that row or column, leaving the rest of the cards untouched. After this operation, Loron determines the ratio of 0s to 1s facing upwards. No matter which row or column Loron chooses, it is *not possible* for this ratio to be

A. $12:13$

B. $2:3$

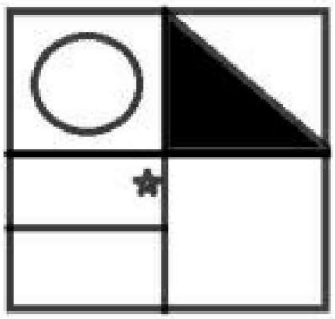
C. $9:16$

D. $3:2$

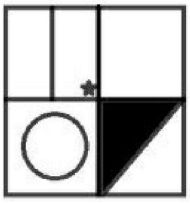
Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Spatial Aptitude

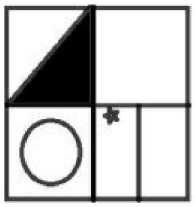
Which of the following figures matches exactly the figure below, but for orientation?



A.



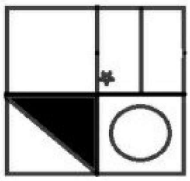
B.



C.



D.



Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

The integers $1, 2, \ldots, n$ are inserted into a Binary Search Tree (BST) in a random order.

What is the **probability** that the **right subtree** of the root contains **exactly** p nodes?

A. $\frac{1}{n}$

B. $\frac{p}{n}$

C. $\frac{n-p}{n}$

D. $\frac{1}{p+1}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Let L be a singly-linked list X and Y be additional pointer variables such that X points to the first element of L and Y points to the last element of L. Which of the following operations cannot be done in time that is bound above by a constant?

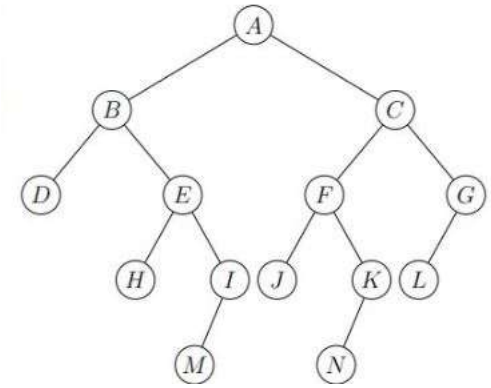
A. Delete the first element of L.
B. Delete the last element of L.
C. Add an element after the last element of L.

D. Add an element before the first element of L.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Suppose that the figure to the right is a binary search tree. The letters indicate the names of the nodes, not the values that are stored. What is the inorder predecessor node, in terms of value, of the root node A?



A. D

B. H

C. I

D. M

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following Python function:

```
def HowManyDash(n):
    if n == 0:
        print('-', end='')
    elif n == 1:
        print('-', end='')
    else:
        HowManyDash(n - 1)
        HowManyDash(n - 2)
```

How many '-' characters are printed when the function call `HowManyDash(10)` is executed?

A. 9

B. 10

C. 55

D. 89

Your Answer: D | Correct Answer: D | Correct | Time taken: 19min 43sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Programming in Python

Consider the following Python function:

```
def fun(A):
    s = 0
    for i in range(len(A)):
        if A[i] % 2 == 0:
            s = s + A[i]
        else:
            s = s - A[i]
    return s
```

If `A = [1, 2, 3, 4, 5]`, what will `fun(A)` return?

A. 3

B. -3

C. 5

D. -5

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Assume we are doing linear regression with Mean-Squared Loss and L2-regularization on four one-dimensional data points. Our prediction model can be written as $f(x) = ax + b$ and the optimization problem can be written as

$$a^\star, b^\star = \underset{a,b}{\operatorname{argmin}} \sum_{n=1}^{4} [y_n - f(x_n)]^2 + \lambda a^2.$$

Assume that our data points are $Y = [1, 3, 2, 4]$ and $X = [-2, -1, 0, 3]$. For example $y_1 = 1$ and $x_1 = -2$. What is the optimal value for the bias, $b^\star$?

A. 2

B. 2.5

C. Depends on the value of λ

D. None of the above answers.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

Let $x_1, x_2, \ldots, x_n$ be d-dimensional data points $(d > n)$ and X be the matrix of shape $d \times n$ containing the data points. The kth largest eigenvalue and corresponding unit eigenvector of $X^T X$ is λ and α_k, respectively. What will be the projection of x_i on the kth principal component?

A. $x_i^T \alpha_k$

B. $\frac{x_i^T \alpha_k}{\sqrt{n\lambda}}$

C. $\frac{x_i^T X \alpha_k}{\sqrt{\lambda}}$

D. $x_i^T X \alpha_k$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Machine Learning

An SVM with hinge loss is trained on a dataset containing $\mathbf{200}$ training samples. After training, the SVM has $\mathbf{40}$ **support vectors.**

What is the **maximum possible value** of the **leave-one-out (LOO) error** for this SVM?

A. 0.10

B. 0.20

C. 0.25

D. 0.40

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose $X \sim \text{Binomial}(10, \frac{1}{2})$ and $Y \sim \text{Binomial}(11, \frac{1}{2})$, where X and Y are independent random variables. Then $P(X < Y)$ is

A. less than $\frac{1}{2}$

B. equal to $\frac{1}{2}$

C. greater than $\frac{1}{2}$ but less than or equal to $\frac{10}{11}$

D. greater than $\frac{10}{11}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Suppose that X is a random variable such that

$$P(X \in \{0, 1, 2\}) = 1.$$

If for some constant c,

$$P(X = i) = c\, P(X = i - 1), \quad i = 1, 2,$$

then the value of $E[X]$ is

A. $\frac{1}{1+c+c^2}$

B. $\frac{c+2c^2}{1+c+c^2}$

C. $\frac{c+c^2}{1+2c}$

D. $\frac{3c}{1+2c}$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 1 | Penalty: 0 | Probability

A discrete random variable X has PMF as given below where a, b, c are constants.

x	1	2	3	4
$P(X=x)$	a	b	c	0.3

The CDF $F_X(x)$ is given below

x	1	2	3	4
$F_X(x)$	0.2	0.6	0.7	d

Find the value of $a+b+c+d$.

Your Answer: | Correct Answer: 1.7 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Probability

Let X_1, X_2, X_3, X_4, X_5 be i.i.d. random variables having a continuous distribution function. Then

$$P\big(X_1 > X_2 > X_3 > X_4 > X_5 \mid X_1 = \max(X_1, X_2, X_3, X_4, X_5)\big)$$

is

A. $1/4$

B. $1/5$

C. $1/4!$

D. $1/5!$

Your Answer: C | Correct Answer: C | Correct | Time taken: 02min 50sec | Discuss

Q #13 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let W be a vector space over $\mathbb{R}$ and let $T : \mathbb{R}^6 \to W$ be a linear transformation such that

$$S = \{Te_2, Te_4, Te_6\}$$

spans W.

Which one of the following must be **TRUE**?

A. S is a basis of W

B. There exists a vector $y \in W$ such that $y \neq T(x)$ for all $x \in \mathbb{R}^6$

C. $\{Te_1, Te_3, Te_5\}$ spans W

D. There exists a nonzero vector $x \in \mathbb{R}^6$ such that $T(x) = 0$

Your Answer: C | Correct Answer: D | Incorrect | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be an $n \times m$ matrix and b be an $n \times 1$ vector (with real entries). Suppose the equation

$$Ax = b, \quad x \in \mathbb{R}^m$$

admits a unique solution. Then we can conclude that

A. $m > n$

B. $n \geq m$

C. $n = m$

D. $n > m$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

For what values of r is the vector $(3, 2, r, 0)$ in $\mathbb{R}^4$ contained in the subspace generated by $(1, 0, 0, 0), (0, 1, 2, 0)$ and $(0, 1, 1, 1)$?

A. For no value of r.

B. For exactly one value of r.

C. For more than one but finitely many values of r.

D. For infinitely many values of r.

Your Answer: B | Correct Answer: B | Correct | Time taken: 03min 24sec | Discuss

Q #16 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let $q_1, q_2 \in \mathbb{R}^n$ be vectors such that

$$\|q_1\| = \|q_2\| = 1 \quad \text{and} \quad q_1^T q_2 = 0.$$

Which of the following is the **projection matrix** onto $\text{span}\{q_1, q_2\}$?

A. $q_1 q_1^T + q_2 q_2^T$

B. $q_1^T q_1 + q_2^T q_2$

C. $q_1 q_2^T + q_2 q_1^T$

D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 1 | Penalty: 0 | Linear Algebra

Consider vectors in the two-dimensional real plane $\mathbb{R}^2$ (the xy-plane).

Suppose we choose a collection of non-zero vectors with the property that the dot product of every distinct pair of vectors in the collection is strictly negative. That is, for any two different vectors $\mathbf{u}$ and $\mathbf{v}$ in the collection,

$$\mathbf{u} \cdot \mathbf{v} < 0.$$

What is the largest possible number of vectors that can be chosen in the xy-plane such that the dot product of any two of them is strictly negative?

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Linear Algebra

Let A be a real, symmetric, square matrix of dimension $2n$. Which of the following statements correctly describes the eigenvalues of A?

A. A has $2n$ distinct real eigenvalues.
B. A has $2n$ real eigenvalues, not necessarily distinct.
C. The eigenvalues of A occur as n distinct pairs of complex conjugate numbers.
D. The eigenvalues of A occur as n pairs of complex conjugate numbers, not necessarily distinct.

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Let x be a variable that takes real values, and let $f(x) = x^3 - 3x$. Which of the following statements is/are true?

A. $f(x)$ has a local maximum at $x = -\sqrt{3}$

B. $f(x)$ has a local maximum at $x = -1$

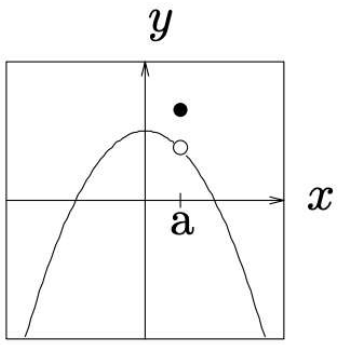
C. $f(x)$ has a local minimum at $x = \sqrt{3}$

D. $f(x)$ has a global minimum at $x = 1$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Select Type | Award: 1 | Penalty: 0 | Calculus

Let f be a function whose graph is shown below.



Then which of the following statements are **correct**?

A. $\lim_{x \to a} f(x)$ exists.

B. $f(a)$ exists.

C. f is continuous at $x = a$.

D. $\lim_{x \to a} f(x) = f(a)$.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

Let $\mathrm{R}_1, \mathrm{R}_2, \ldots, R_n$ be a decomposition of schema U. Let $u(\mathrm{U})$ be a relation, and let $r_i = \Pi_{\mathrm{R}_i}(u)$

Which of the following is true?

A. $u \subseteq r_1 \bowtie r_2 \bowtie r_3 \bowtie r_4 \bowtie \ldots r_n$

B. $u \supseteq r_1 \bowtie r_2 \bowtie r_3 \bowtie r_4 \bowtie \ldots r_n$

C. $u = r_1 \bowtie r_2 \bowtie r_3 \bowtie r_4 \bowtie . . r_n$

D. None of the above

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Databases

A weak entity

A. inherits part of its key from the "parent" entities to which it is related.

B. must have partial participation in its identifying relationship set.

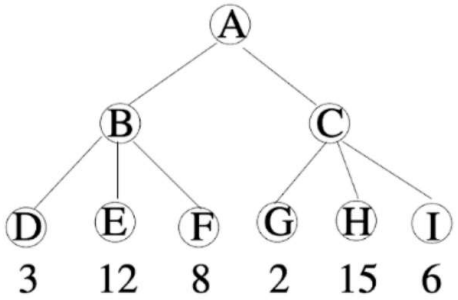
C. is an entity with no attributes besides its key.

D. is never mapped to a table in conversion to a relational schema.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Which nodes will be pruned by alpha-beta pruning, assuming that the root node A is a **MAX** node, its children B and C are **MIN** nodes, and the game tree is evaluated strictly from **left to right**?



A. I

B. HI

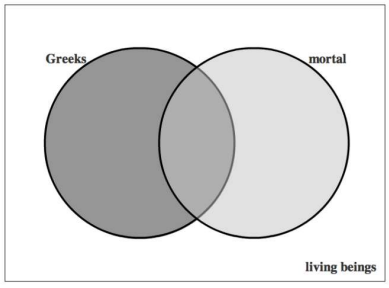
C. GHI

D. none of the above

Your Answer: C | Correct Answer: B | Incorrect | Time taken: 02min 09sec | Discuss

Q #24 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Artificial Intelligence

Consider the following Venn diagram:



Which one of the following categorical propositions can be used in order to express that the dark grey area on the left of the diagram does not contain any elements?

A. Some Greeks are not mortal

B. Some Greeks are mortal

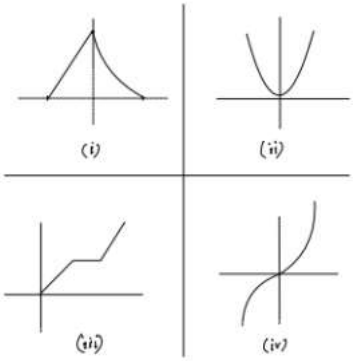
C. All Greeks are mortal

D. No Greeks are mortal

Your Answer: A | Correct Answer: C | Incorrect | Time taken: 00min 52sec | Discuss

Q #25 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Calculus

Which of the following plots correspond to a function whose derivative is continuous?



A. (i) and (ii)

B. (ii) and (iii)

C. (iii) and (iv)

D. (ii) and (iv)

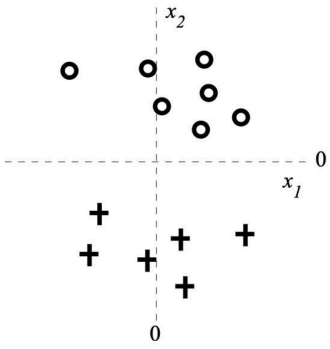
Your Answer: A | Correct Answer: D | Incorrect | Time taken: 01min 48sec | Discuss

Q #26 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

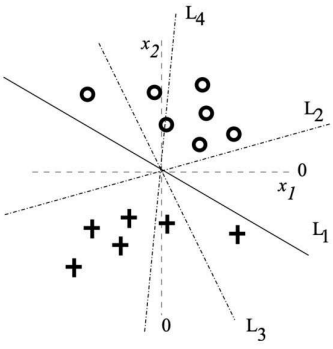
We consider a binary classification problem solved using logistic regression (without bias term):

$$\hat{P}(y = 1 \mid \mathbf{x}, w_1, w_2) = \frac{1}{1 + \exp(-w_1 x_1 - w_2 x_2)}.$$

The training data is linearly separable with zero training error. The dataset along with candidate decision boundaries is shown below.



(a) The 2-dimensional data set



(b) The points can be separated by L_1 (solid line). Possible other decision boundaries are shown by L_2, L_3, L_4.

Consider the following regularized objective:

$$\sum_{i=1}^{n} \log p(y_i \mid x_i, w_1, w_2) \; - \; \frac{C}{2} w_2^2,$$

where **only** w_2 is penalized and C is large. Which of the following decision boundaries $(L2, L3, L4)$ shown in the figure can arise as a result of this regularization?

Select all that apply.

A. $L2$

B. $L3$

C. $L4$

D. None of the decision boundaries shown are consistent with penalizing only w_2

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let X be a uniformly distributed random variable with mean $\mu_X = 15$ and variance $\sigma_X^2 = \frac{25}{3}$. Calculate $P(X > 17)$.

Your Answer: 0.3 | Correct Answer: 0.3 | Correct | Time taken: 16min 44sec | Discuss

Q #28 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Suppose (X, Y) follows a bivariate normal distribution with means μ_1, μ_2, standard deviations σ_1, σ_2, and correlation coefficient ρ, where all the parameters are unknown.

Then testing the null hypothesis

$$H_0 : \sigma_1 = \sigma_2$$

is equivalent to testing the independence of:

A. X and Y

B. X and $X - Y$

C. $X + Y$ and Y

D. $X + Y$ and $X - Y$

Your Answer: A | Correct Answer: D | Incorrect | Time taken: 04min 53sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let A be an $n \times n$ real symmetric non-singular matrix. Suppose there exists $x \in \mathbb{R}^n$ such that $x^T A x < 0$. Then we can conclude that

A. $\det(A) < 0$

B. $B = -A$ is positive definite

C. $\exists\, y \in \mathbb{R}^n : y^T A^{-1} y < 0$

D. $\forall\, y \in \mathbb{R}^n : y^T A^{-1} y < 0$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Consider a hash table with N slots. It is given that the collision resolution technique used in chaining. Assume simple uniform hashing, what is the probability that the last k slots are unfilled after the first 'r' insertions?

A. $\left(1 - \frac{N}{k}\right)^r$

B. $\left(1 - \frac{k}{N}\right)^r$

C. $\left(1 + \frac{N}{k}\right)^{r-1}$

D. $\left(1 - \frac{k}{N}\right)^{r-1}$

Your Answer: B | Correct Answer: B | Correct | Time taken: 03min 27sec | Discuss

Q #31 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following Python function that compares two lists of numbers and checks whether they contain exactly the same set of elements (with no duplicates).

```
def listcmp(list1, list2):
    for num in list1:
        if num not in list2:
            return False
        else:
            # Remove the first occurrence of num from list2
            list2.remove(num)
    # Return True if and only if list2 is now empty
    return list2 == []
```

The Python method `list.remove(x)` scans the list from the beginning to find `x` and removes its first occurrence.

Let n be the length of `list1` and m be the length of `list2`.

What is the worst-case time complexity of the above implementation?

A. $\Theta(n + m)$

B. $\Theta(n \log m)$

C. $\Theta(nm)$

D. $\Theta(m^2)$

Your Answer: C | Correct Answer: C | Correct | Time taken: 04min 00sec | Discuss

Q #32 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Let G be a connected simple graph that **contains no odd-length cycles**, and all its edge weights are distinct. Which of the following statements about the minimum spanning tree (MST) of G is/are **NOT** necessarily true?

A. Every MST of G contains the lightest edge.

B. Every MST of G contains the second lightest edge.

C. Every MST of G contains the third lightest edge.

D. The heaviest edge of G is excluded from every MST of G.

Your Answer: A;B;C | Correct Answer: D | Incorrect | Time taken: 00min 01sec | Discuss

Q #33 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the initial array: $[6, 2, 7, 8, 3, 9, 5, 10, 1, 4]$.

We have applied different sorting algorithms to this array. The arrays shown below (left column) represent intermediate steps (snapshots) of a sort in progress. Each sorting algorithm is used exactly once.

Additional information:

- If the sorting algorithm is quick sort, the first element of the array is chosen as the pivot.
- If the sorting algorithm contains multiple loops, the array shown is after a few complete passes of the outermost loop.

Intermediate State of array	Sorting algorithm
P. [1, 2, 3, 8, 7, 9, 5, 10, 6, 4]	i. Bubble sort
Q. [2, 3, 6, 5, 7, 1, 4, 8, 9, 10]	ii. Selection sort
R. [4, 2, 1, 5, 3, 6, 8, 10, 7, 9]	iii. Insertion sort
S. [2, 3, 6, 7, 8, 1, 4, 5, 9, 10]	iv. Merge sort
T. [2, 6, 7, 8, 3, 9, 5, 10, 1, 4]	v. Quick sort

Match the intermediate states of the array (from the left column) to the sorting algorithms (in the right column) that could have produced them.

A. P-ii; Q-i; R-v; S-iv; T-iii

B. P-ii; Q-iv; R-v; S-i; T-iii

C. P-i; Q-ii; R-v; S-iv; T-iii

D. P-ii; Q-i; R-iv; S-v; T-iii

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #34 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Programming in Python

Consider the following Python code:

```
def f(n):
    if n == 0:
        return [0]
    L = f(n - 1)
    return [x + n for x in L] + [n]
```

What will be the output of `f(3)`?

A. `[6, 6, 5, 3]`

B. `[6, 5, 4, 3]`

C. `[3, 4, 5, 6]`

D. `[3, 3, 2, 1]`

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #35 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider data points $x_1, x_2, \ldots, x_n \in \mathbb{R}^d$. At iteration t, each point x_i is assigned to a cluster indexed by $c_i^{(t)}$, with corresponding cluster mean $\mu_j^{(t)}$. After reassignment in the next iteration, the cluster index of x_i becomes $c_i^{(t+1)}$, and the updated cluster means are $\mu_j^{(t+1)}$.

Consider the following three quantities:

(1) $\sum_{i=1}^{n} \left\| x_i - \mu_{c_i^{(t+1)}}^{(t)} \right\|^2$

(2) $\sum_{i=1}^{n} \left\| x_i - \mu_{c_i^{(t)}}^{(t)} \right\|^2$

(3) $\sum_{i=1}^{n} \left\| x_i - \mu_{c_i^{(t+1)}}^{(t+1)} \right\|^2$

Which of the following relationships among (1), (2), and (3) is correct?

A. $(1) > (2) > (3)$

B. $(1) < (2) < (3)$

C. $(3) < (1) < (2)$

D. $(2) < (3) < (1)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

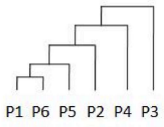
Q #36 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Consider the similarity matrix given below.

	P1	P2	P3	P4	P5	P6
P1	1.00	0.70	0.65	0.40	0.20	0.05
P2	0.70	1.00	0.95	0.70	0.50	0.35
P3	0.65	0.95	1.00	0.75	0.55	0.40
P4	0.40	0.70	0.75	1.00	0.80	0.65
P5	0.20	0.50	0.55	0.80	1.00	0.85
P6	0.05	0.35	0.40	0.65	0.85	1.00

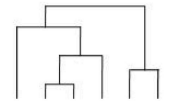
Show the hierarchy of clustering created by the **single-link clustering** algorithm.

A.



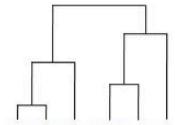
(a)

B.



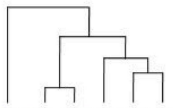
(b)

C.



(c)

D.



(d)

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #37 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose we have a set $S = \{(x_1, y_1), \ldots, (x_m, y_m)\}$ of labeled examples in $\mathbb{R}^n$. Recall that SVM solves the optimization problem

$$\begin{aligned} \min_w \quad & \|w\|_2^2 + C\sum_{i=1}^m \xi_i \\ \text{subject to} \quad & y_i\left(w^T x_i\right) \geq 1 - \xi_i \quad \forall i \\ & \xi_i \geq 0 \quad \forall i \end{aligned}$$

What two quantities is the optimization objective trading off?

A. Maximizing the margin and minimizing the number of misclassifications

B. Maximizing the margin and minimizing how far misclassified points are from the correct side of the margin

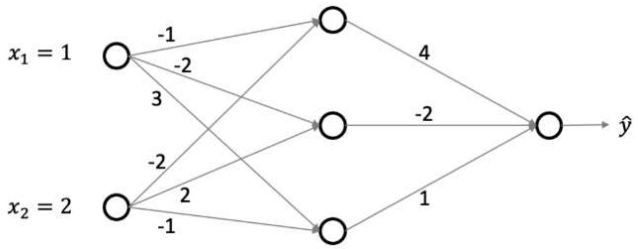
C. Maximizing the margin and minimizing the number of support vectors

D. Maximizing the margin and minimizing the number of nonzero elements in w

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #38 | Numerical Type | Award: 2 | Penalty: 0 | Machine Learning

Consider the following neural network with weights shown in the image below. Every hidden neuron uses the ReLU activation function, and there is no activation function on the output neuron. Assume there are no bias terms. What is the output of this network with the input $x = (1, 2)$? Give a numerical answer.



Your Answer: 0 | Correct Answer: -3 | Incorrect | Time taken: 07min 04sec | Discuss

Q #39 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Machine Learning

Suppose we are training a Naive Bayes classifier to distinguish between students taking $\mathrm{Course\ A}$ and $\mathrm{Course\ B}$ using 12 real-valued features.

We train two models:

- Model M_1 is trained using data from students across both courses in the $\mathrm{F23}$ semester. In this semester, there are G_1 students from $\mathrm{Course\ A}$ and U_1 students from $\mathrm{Course\ B}$.
- Model M_2 is trained using data from the $\mathrm{S23}$ semester. In this semester, there are G_2 students from $\mathrm{Course\ A}$ and U_2 students from $\mathrm{Course\ B}$.

Surprisingly, both M_1 and M_2 learn exactly the same values for all the feature means and variances.

Now suppose a new student, who was not used in training, is classified by both models:

- M_1 predicts that the student is in $\mathrm{Course\ A}$.
- M_2 predicts that the student is in $\mathrm{Course\ B}$.

Given this information, which of the following relationships must be true?

A. $G_1 > G_2$

B. $U_2 > U_1$

C. $G_1 + U_1 > G_2 + U_2$

D. $\dfrac{G_1}{G_2} > \dfrac{U_1}{U_2}$

Your Answer: D | Correct Answer: D | Correct | Time taken: 08min 31sec | Discuss

Q #40 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

Let $X_1, X_2, \ldots$ be a sequence of independent normally distributed random variables with mean 1 and variance 1. Let N be a Poisson random variable with mean 2, independent of $X_1, X_2, \ldots$.

Then, the variance of

$$X_1 + X_2 + \cdots + X_{N+1}$$

is:

A. 3

B. 4

C. 5

D. 9

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #41 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Probability

In a random experiment, a fair coin is tossed once. Then, an unbiased six-faced die is rolled N times, where

$$N = \begin{cases} 100, & \text{if Head appears,} \\ 101, & \text{if Tail appears.} \end{cases}$$

Let Y denote the total number of times 6 appears out of N. Then $P(\text{Head} \mid Y = 15)$ equals:

A. $\frac{516}{1021}$

B. $\frac{505}{1021}$

C. $\frac{201}{1021}$

D. $\frac{1000}{1021}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #42 | Numerical Type | Award: 2 | Penalty: 0 | Probability

Let X and Y be two random variables with joint PMF $f_{X,Y}(x, y)$ given below. Calculate covariance between X and Y.

Y \ X	1	2	3
1	0.25	0.25	0
2	0	0.25	0.25

Your Answer: | Correct Answer: 0.25 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #43 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Consider events $B_1, B_2, \ldots, B_k$ that form a partition of the sample space S. For $i = 1, \ldots, k$, let $\Pr(B_i)$ denote the prior probability of B_i. There is another event A such that $\Pr(A) > 0$. Let $\Pr(B_i \mid A)$ denote the posterior probability of B_i given that the event A has occurred.

Suppose that

$$\Pr(B_1 \mid A) < \Pr(B_1).$$

Which of the following statements are correct? (Multiple options may be correct.)

A. $\sum_{i=1}^{k} \Pr(B_i) = 1$

B. $\sum_{i=1}^{k} \Pr(B_i \mid A) = 1$

C. There exists at least one value of i, $i = 2, \ldots, k$, such that $\Pr(B_i \mid A) > \Pr(B_i)$

D. For all $i = 2, \ldots, k$, $\Pr(B_i \mid A) \leq \Pr(B_i)$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #44 | Multiple Select Type | Award: 2 | Penalty: 0 | Probability

Let X be a discrete random variable with support

$$S = \{-1, 0, 1\}$$

and

$$P(X = 0) = \frac{1}{3}.$$

Which of the following statements are true?

A. $E(X) \leq \frac{2}{3}$

B. $E(X^2) = \frac{2}{3}$

C. $E(|X|) = \frac{2}{3}$

D. $\mathrm{Var}(X) > \frac{2}{3}$

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #45 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let W_1, W_2, W_3 be three distinct subspaces of $\mathbb{R}^{10}$ such that each W_i has dimension 9. Let

$$W = W_1 \cap W_2 \cap W_3.$$

Then we can conclude that—

A. W may not be a subspace of $\mathbb{R}^{10}$.

B. $\dim(W) \leq 8$.

C. $\dim(W) \geq 7$.

D. $\dim(W) \leq 3$.

Your Answer: | Correct Answer: B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #46 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Let $f : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(v, w) = w^T A v$.

Pick the correct statements from below.

A. There exists an eigenvector v of A such that $Av \perp v$.

B. The set $\{ v \in \mathbb{R}^2 \mid f(v, v) = 0 \}$ is a non-zero subspace of $\mathbb{R}^2$.

C. If $v, w \in \mathbb{R}^2$ are non-zero vectors such that $f(v, w) = f(w, w)$, then v is a scalar multiple of w.

D. For every $v \in \mathbb{R}^2$, there exists a non-zero $w \in \mathbb{R}^2$ such that $f(v, w) = 0$.

Your Answer: D | Correct Answer: D | Correct | Time taken: 07min 39sec | Discuss

Q #47 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Linear Algebra

Let V be an 11-dimensional real vector space with ordered basis

$$\{v_0, v_1, v_2, \ldots, v_{10}\}.$$

Define a linear transformation $T : V \to V$ by

$$T(v_0) = 0, \quad T(v_k) = v_{k-1} \text{ for } k = 1, 2, \ldots, 10.$$

Visually, you can think of T as taking v_{10} to v_9, and so on: $v_{10} \mapsto v_9 \mapsto v_8 \mapsto \cdots \mapsto v_1 \mapsto v_0 \mapsto 0$.

Let A be the matrix of T with respect to this basis. Then which of the following statements is correct?

A. $\mathrm{Tr}(A) = 1$

B. $\det(A) = 0$

C. There is no $m \in \mathbb{N}$ such that $A^m = 0$

D. A has a non-zero eigenvalue

Your Answer: B | Correct Answer: B | Correct | Time taken: 04min 40sec | Discuss

Q #48 | Multiple Select Type | Award: 2 | Penalty: 0 | Linear Algebra

Let $A \in \mathbb{R}^{m \times n}$ be a matrix of rank r with Singular Value Decomposition

$$A = U \Sigma V^T$$

Assume that $\mathrm{rank}(A) = n$ (i.e., A has full column rank).

Define $B = (A^T A)^{-1} A^T$.

Which of the following statements are **correct**?

A. Since $r = n$, the matrix V is square and satisfies $(V^T)^{-1} = V^T$.

B. The matrix A^TA can be written as $A^TA = V\Sigma^2V^T$.

C. The matrix B can be expressed as $B = V\Sigma^{-1}U^T$.

D. If $r = n = m$, then $B = A^{-1}$.

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #49 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let f be a single variable, real valued function such that all its derivatives exist. We say that f changes sign at some point x_0 in the domain if the product $f(x_0 + \epsilon) \cdot f(x_0 - \epsilon) < 0$ for small enough, positive ϵ.

- A point x_0 in the domain is called a turning point of f, if the first derivative f' changes sign at x_0.
- A point x_0 in the domain is called flex point of f, if the second derivative f'' changes sign at x_0.

Which of the following statements is/are true?

A. If x_0 is a flex point of f then $f''(x_0) = 0$.

B. If $f'(x_0) = 0$ then x_0 is a turning point of f.

C. If $f''(x_0) = 0$ then x_0 is a flex point of f.

D. If x_0 is a flex point of f then $f'(x_0) \neq 0$.

Your Answer: A;B;C | Correct Answer: A | Incorrect | Time taken: 10min 13sec | Discuss

Q #50 | Multiple Select Type | Award: 2 | Penalty: 0 | Calculus

Let $f(x)$ be a real-valued function all of whose derivatives exist. Recall that a point x_0 in the domain is called an inflection point of $f(x)$ if the second derivative $f''(x)$ changes sign at x_0. Given the function $f(x) = \frac{x^5}{20} - \frac{x^4}{2} + 3x + 1$, which of the following statements are true?

A. $x_0 = 0$ is not an inflection point

B. $x_0 = 6$ is the only inflection point

C. $x_0 = 0$ and $x_0 = 6$, both are inflection points

D. The function does not have an inflection point

Your Answer: A;B | Correct Answer: A;B | Correct | Time taken: 17min 51sec | Discuss

Q #51 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Databases

Consider the following database tables used to track doctor visits to animals in a zoo, where the primary keys are shown underlined. The field "aid" of the table "visits" is a foreign key referring to "aid" of the table "animals". The field "did" of the table "visits" is a foreign key referring to "did" of the table "doctors".

Each row of the animals table describes a distinct animal. Each row of the doctors table describes a distinct doctor at the zoo. Each row of the visits table describes a distinct visit of an animal to a doctor.

(a) animals

aid	animal_type	name	age	color
0	rabbit	Bugs	2	white
1	bear	Air	5	golden
2	bear	Care	1	golden
3	cat	Grumpy	75	gray

(b) doctors

did	name
0	Turk
1	House
2	Dre
3	Bailey

(c) visits

vid	aid	did
132	1	0
145	2	1
167	0	3
168	2	3
169	2	2

Which of the following SQL queries computes the names of the animals in our zoo who visited the doctor more than 2 times?

A. **SELECT** name **FROM** animals, visits
WHERE COUNT(*) > 2
GROUP BY animals.aid, animals.name;

B. **SELECT** name **FROM** animals, visits
GROUP BY animals.aid, animals.name
HAVING COUNT(*) > 2;

C. **SELECT** name **FROM** animals, visits
WHERE visits.aid = animals.aid **AND** COUNT(*) > 2
GROUP BY animals.aid, animals.name

D. **SELECT** name **FROM** animals, visits
WHERE visits.aid = animals.aid
GROUP BY animals.aid, animals.name
HAVING COUNT(*) > 2;

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #52 | Numerical Type | Award: 2 | Penalty: 0 | Databases

Let R be the set of attributes of relation instance r and S the set of attributes of relation instance s such that $S \subset R$.

In the relational algebra, consider an operation $\#$, defined as below:
$r\#s$ is defined to be the largest relation q such that $q \times s \subseteq r$.

Instances r, s of the schema R, S, respectively, are given below.

$r(R)$

A	B	C
$a1$	$b1$	$c1$
$a2$	$b1$	$c1$
$a1$	$b2$	$c1$
$a1$	$b2$	$c2$
$a2$	$b1$	$c2$
$a1$	$b2$	$c3$
$a1$	$b2$	$c4$
$a1$	$b1$	$c5$

$s(S)$

C
$c1$
$c2$

The number of rows in $r\#s$?

Your Answer: | Correct Answer: 2 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #53 | Multiple Select Type | Award: 2 | Penalty: 0 | Databases

Let $R(A, B, C, D, E)$ be a relation schema with the following Functional Dependencies (FDs):

$$F = \{AB \to C, \quad C \to D, \quad D \to E, \quad E \to A\}$$

Which of the following statements are CORRECT?

A. The decomposition of R into $R_1(A, B, C)$ and $R_2(C, D, E)$ is Lossless.

B. The relation R is in 3NF but not in BCNF.

C. The attribute B is a prime attribute.

D. The decomposition of R into $R_1(A, B, C)$, $R_2(C, D)$, $R_3(D, E)$, $R_4(E, A)$ is Dependency Preserving.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #54 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

A Bayes network has five variables, with a topological ordering v, w, x, y, z. All variables are binary, except variables v and y, which are ternary. With the given topological ordering, the following conditional independencies hold in the network:

$$\{v, w\} \perp y \mid x \qquad \{v, x\} \perp z \mid \{w, y\}$$

Derive the factorization of the joint distribution corresponding to this Bayes network. **How many parameters does the Bayes network in question have?**

A. 24

B. 18

C. 21

D. 23

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #55 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Artificial Intelligence

"Every cat loves its mother or father" can be translated as:

A. $\forall x. \neg \text{Cat}(x) \vee \text{Loves}(x, \text{mother}(x)) \vee \text{Loves}(x, \text{father}(x))$

B. $\forall x. Cat(x) \to \text{Loves}(x, \text{mother}(x)) \vee \text{father}(x))$

C. $\forall x. \text{Cat}(x) \wedge (\text{Loves}(x, \text{mother}(x)) \vee \text{Loves}(x, \text{father}(x)))$

D. $\forall \text{x}. \text{Cat}(x) \vee (\text{Loves}(x, \text{mother}(x)) \vee \text{Loves}(x, \text{father}(x)))$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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